Half of the streamlines built in a whole human brain tractogram is anatomically uninterpretable.

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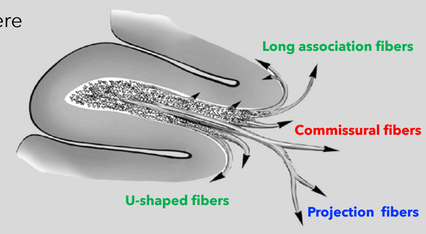
Introduction

- One of the strategies to study human white matter (WM) with diffusion imaging and tractography is to reconstruct millions of streamlines using whole brain fiber tracking algorithms, then followed by advanced filtering or manual dissection with regions-of-interest from a segmentation tool (e.g. FreeSurfer or JHU template) to extract specific bundles [1].
- Despite the developments of algorithms with anatomical constraints to prevent false-positive [2,3], the whole brain tractograms still include a large number of anatomically implausible streamlines.

In this work, we estimate the plausible/implausible streamlines ratio of a whole human brain tractogram.

What is a plausible streamline ?

- Association fibers
  both terminations in the CGM/SWM within the same hemisphere
- Commissural fibers
  one termination in CGM/SWM in each hemisphere
- Projection fibers
  one end in the brainstem or the striatum or the thalamus and the other in CGM/SWM in the same hemisphere

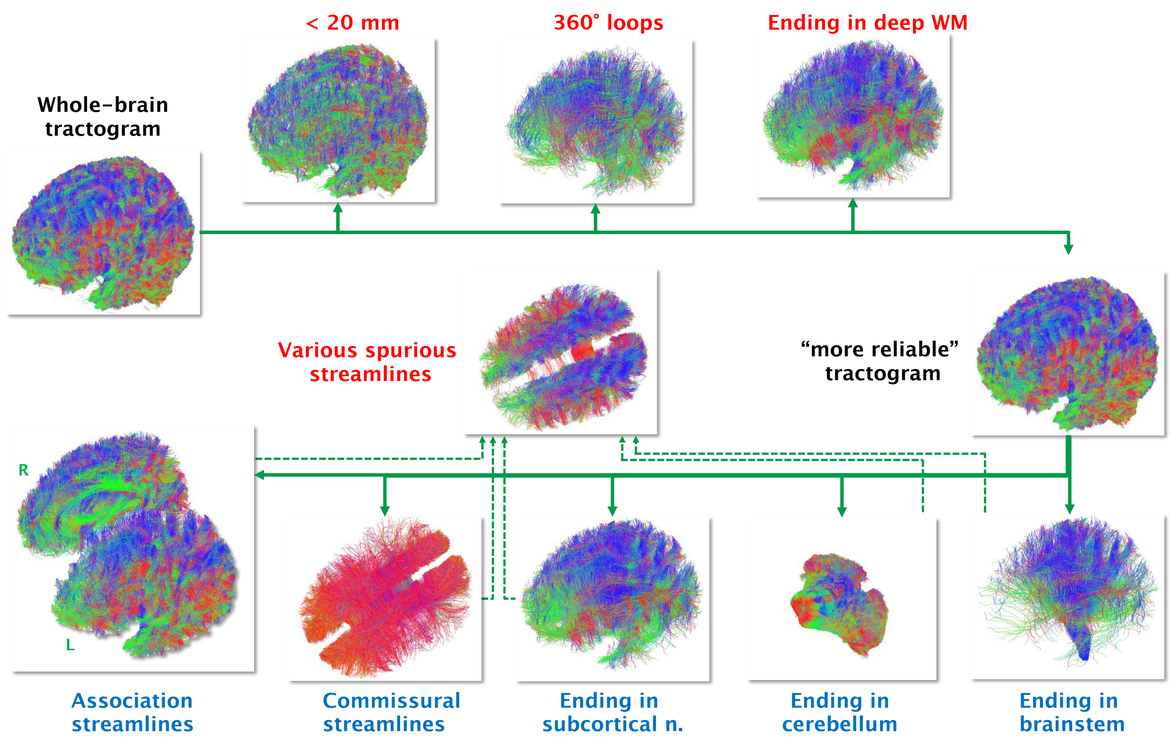


What is an implausible streamline ?

- Too short streamlines for a reliable anatomical labelling (< 20 mm);
- Streamlines with 360°-loops;
- Broken streamlines, *i.e.* with at least one termination in the deep WM;
- Spurious streamlines from association, commissural and projection ones.

ExTractor

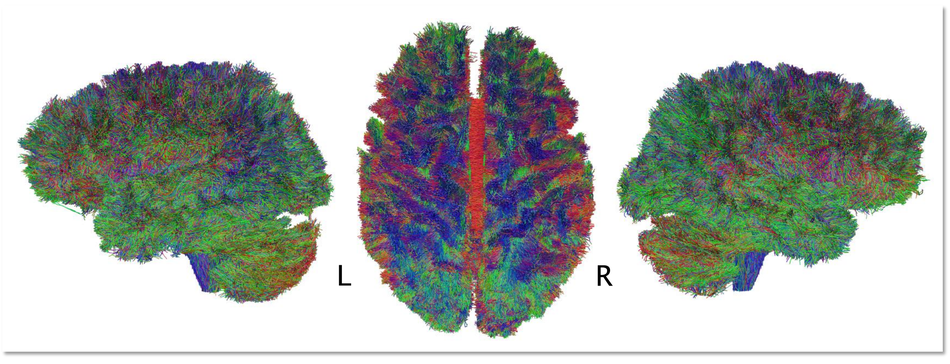
- A tool for sorting plausible and implausible streamlines from a whole brain tractogram before any additional extraction of specific bundles or structural connectivity analyses.
- Based on JHU template with 176 anatomical structures delineating cortical grey (GM,■), superficial white (SWM,■) and deep white matter (DWM,■) [7].



Methods

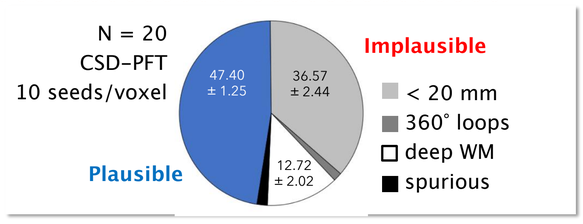
Diffusion-weighted imaging and standard tractography

- 20 healthy individuals of the BIL&GIN database [4] and 20 healthy individuals of the HCP database [5] ;
- T1-weighted and 2 x 42 3D DWI volumes (b=1000s/mm², 2 mm isotropic voxel);
- Standard individual whole-brain tractograms were computed on fODF using an advanced streamline probabilistic T1-weighted anatomically-constrained particle filtering tractography (CSD-PFT, [6]).
  - 10 seeds/voxel of the GM/WM interface
  - Ending in the weighted-GM map
  - Rejecting streamlines ending in the CSF
  - Limiting the streamline length [10-250mm]

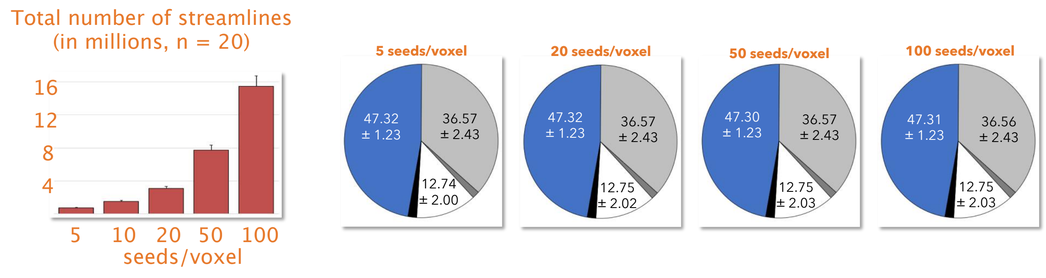


Results

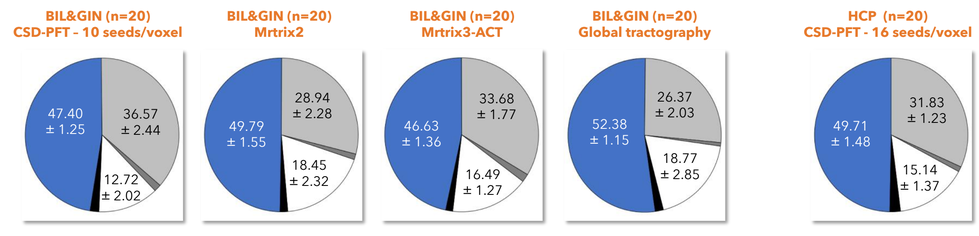
- Half of the total number of streamlines were anatomically implausible.



- Invariant to the number of seeds.



- Invariant to the tractography algorithm or the quality of the DWI data.



Conclusion

- While tractography algorithms are currently challenged by important issues that have to be addressed to improve human brain connectivity mapping [8], we demonstrate that the amount of implausible streamlines in typical whole brain tractograms is far to be negligible and needs to be removed from the data before any structural connectome building [10].
- A better knowledge of these invalid streamlines, notably their profile, their distribution and frequency across the whole brain, will foster the development of future tractography methods that will avoid their building during the fiber-tracking process.

1. Jeurissen et al. (2019) NMR Biomed 32:e3785; 2. Smith et al. (2012) NeuroImage 62:1924-1938; 3. Girard et al. (2014) NeuroImage 98:266-278; 4. Mazoyer et al. (2016) Neuroimage 124B:1225-1231; 5. Van Essen et al. (2013) Neuroimage 80:62-79; 6. Girard et al. (2014) Neuroimage 98:266-278; 7. Oishi et al. (2009) Neuroimage 46:486-499; Maier-Hein et al. (2017) Nature Communication 8:1349; 10. Zhang et al. (2018) NeuroImage, 2018. 172:130-145.

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